IMA EDUCATIONAL Case Journal | Case Study

ima® The Association of Accountants and Financial Professionals in Business

ISSN 1940-204X

À Votre Santé: Product Costing and Decision Analysis in the Wine Industry

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BACKGROUND

The Aproveche family owns and operates a small independent winery located in the Napa Valley California American Viticultural Area (AVA).[1] À Votre Santé (AVS), which means "to your health" in French, enjoys a reputation for producing small amounts of quality wines. AVS was started by Jerome Aproveche in 2005 as an extension of the family's grape-growing operations and as a means to involve his children in the wine business. This path is not an unusual one in Napa Valley, where there are more than 600 grape growers and over 500 wine producers and blenders, many of which are small family-owned operations. In the Napa Valley region, over 9 million cases of wine are typically produced and sold annually. With its operation of about 4,000 cases produced each year, AVS is considered to be a small winery.

AVS is managed by Kay Aproveche, the founder's daughter. AVS buys two types of wine grapes from the family's grape-growing operations – a Chardonnay and a generic white grape. AVS is responsible for harvesting and processing the grapes into three types of wine that have been made by AVS since its founding – a Chardonnay-Estate, a Chardonnay, and a Blanc de Blanc. AVS wines are bought by local restaurants to serve to their customers; every year AVS has sold all the wine that it has produced. In 2010, AVS earned an 11.1% profit margin on sales of $848,000 (see Exhibit 1). Kay was concerned that the 2010 profit margin had declined from the 2009 profit margin of 14%, despite a small increase in sales volume.

[1] AVAs are wine grape-growing areas established by the Department of the Treasury's Alcohol and Tobacco Tax Trade Bureau. AVAs are established to help vintners better describe the origin of their wines. There are currently about 200 AVAs in the United States; the Napa Valley AVA consists of 15 AVAs, each determined by geographical boundaries.

WINEMAKING PROCESS

The process of winemaking is fairly simple, yet requires much attention to process details. Harvest takes place in the late summer and early fall months; typically, the time elapsed from harvest to final sale is about 11 months. After harvest, the grapes are brought to the winery for washing and crushing. The crushing process separates the juice from the pulp, skin, and stems. The juice is used to make the wine; the pulp, skin, and stems are recycled back onto the fields whenever possible or disposed of. The amount of grapes available for harvest and the amount of juice generated from the grapes is dependent each year on a number of climatic and growing factors such as temperature, length of growing season, rootstock, and fertilizers used.

Once the juice is extracted, it moves into the fermenting process. The Chardonnay wine grape is fermented using oak barrels; the oak in the barrels gives flavor to the Chardonnay wine. The generic white grape juices are fermented in a steel holding tank rather than in barrels. All fermenting takes place in a temperature-controlled environment; however, each fermenting method results in some wine loss through evaporation.

PRODUCT INFORMATION

Data related to the three AVS wines is as follows:

- Chardonnay-Estate contains only Chardonnay grapes that are grown for AVS at the family vineyards. The 2010 sales projection for Chardonnay-Estate wine was estimated to be 24,000 bottles at an average sales price of $22 per bottle.
- Regular Chardonnay is blended by combining the Chardonnay wine left over after bottling the Chardonnay-Estate with the fermented generic wine; the blend mixture is two parts Chardonnay grapes and one part generic grapes. The average sales price is $16/bottle.
- Blanc de Blanc wine is made from all remaining generic white grapes. The average sales price is $11/bottle.

All three wines are bottled at AVS using one bottling line. In a typical year, AVS bottles enough Chardonnay-Estate to meet the sales projection, and then bottles the regular Chardonnay after blending all remaining Chardonnay wine with the necessary amount of generic grapes. The Blanc de Blanc is the last wine to be bottled, using all remaining generic white grapes.

ADDITIONAL OPERATIONAL AND COST DATA

CHARDONNAY GRAPES

- 2009 harvest: 100,000 pounds
- purchase price: $85,500
- expected loss in volume through fermentation: 10%

GENERIC WHITE GRAPES

- 2009 harvest: 60,000 pounds
- purchase price: $38,500
- expected loss in volume through fermentation: 5%

WINEMAKING

- Chardonnay grapes are fermented in oak barrels; each barrel results in the production of 40 cases of wine.
- Barrels cost $500 apiece. Because the barrels lose some of their oak flavorings each use, AVS uses the barrels for four years and then sells them to other wineries for $200. AVS's current barrels have been used for four years, so AVS will purchase all new barrels for the 2009 harvest.
- Generic white grapes are fermented in the holding tank; the tank can hold up to the equivalent of 2,000 cases of wine.

BOTTLING

- Requires 36 pounds of grapes (post-fermenting) for one case (12 bottles) of wine
- In the bottling process, the wine is put into bottles, with both corks and labels added during this process. The materials cost associated with the bottles, corks, and labels averages $2.50/bottle.

DIRECT LABOR

- Harvest labor is paid an average of $9.00/hour. An average of 80 pounds of grapes can be harvested each hour.
- Crush labor is paid an average of $10.00/hour. The crushing process typically takes 300 hours to complete.

OVERHEAD EXPENSES

- Administrative rent and office expenses: currently $20,000 annually.
- Depreciation is charged based on the following equipment schedule:

Equipment	Cost	Est. Life
Tractors	$15,000	10 years
Crushers	$6,000	10 years
Holding Tank	$40,000	20 years
Bottle Lines	$10,000	10 years
Other Production Equipment	$15,000	5 years

- Indirect materials – Part of the winemaking process involves introducing yeasts and other additives into the wine to help the fermentation process and to help balance the flavors in the wine. Indirect production materials average $1.55 per case of wine.
- Lab expenses – Lab expenses of $8,000 are incurred for lab supplies and equipment. The lab is used by the production supervisor and the wine master to test the grapes and wine at various stages of production.
- Liquor taxes – AVS is required to pay a liquor excise tax of $3/bottle on every bottle of wine sold.
- Production office – AVS pays a part-time employee to help administer the production function. This employee orders supplies, reviews and approves production invoices, and performs other administrative functions. The production office budget is $12,000 annually.
- Sales and related – Kay's sister, Maria, is paid $30,000/year on a contract basis to sell AVS wines. She works through distributors, who are paid $2/bottle for each bottle sold.
- Supervision – Kay's brother, Albert, supervises the production of wine from the harvest through the bottling processes. His salary and benefits total $55,000 annually.

- Utilities – Utilities costs of $5,500 are incurred primarily to maintain a constant temperature in the fermenting process.
- Waste treatment – After crushing, the pulp, skins, and stems that are left over must be disposed of. One-half of the waste can be recycled back onto the fields as a compost material; the other half must be disposed of at a landfill with a dumping cost of $2,000.
- A winemaster is employed on a contract basis to help formulate and test the wines. AVS pays the winemaster $5,000 for each type of wine that is formulated.
- As the AVS manager, Kay's annual salary and benefits total $75,000.

CASE QUESTIONS

1. **Contribution margin analysis:** Create a single company-wide contribution margin income statement for AVS that includes each expense category. Also, calculate the weighted-average revenue and weighted-average operating margin for one bottle of wine. (Note: Do not break out the variable or the fixed costs by type of wine.)
2. **Decision analysis:** Another grower has available 20,000 pounds of Chardonnay grapes from the 2009 harvest. AVS has the opportunity to buy the juice from these grapes (they have already been harvested and crushed).
   a. If AVS could blend these grapes with the generic white grapes (using the 2:1 blend formula) to produce a new Chardonnay wine to be priced at $14/bottle, and required a 15% return on sales for this wine, what is the maximum amount that AVS would pay for a pound of grapes?
   b. Other than the cost of the grapes, what factors would you consider to support your purchase of the additional grapes, and what factors would cause you to reject buying the grapes?
3. **Product profitability analysis:** Kay would like to be able to better assign costs to each of AVS's products, and has collected the following information about AVS activities.
   - **Harvesting activity –** The Chardonnay grapes can be harvested at the average rate of 71.5 pounds/hour, while the generic white grapes are harvested at the rate of 100 pounds/hour. Harvesting includes costs related to tractors and harvest labor.
   - **Crush activity –** Relates to the pounds of grapes crushed. Includes costs of crusher equipment and crush labor.
   - **Fermenting activities –** Barrel costs relate to the Chardonnay wines by the percentage of wine used in each wine type. Holding tanks relate to the Blanc de Blanc and the regular Chardonnay wines by the percentage of wine used in each wine type.
   - **Lab and supervision activities –** Albert estimates that these activities are split 60% Chardonnay-Estate, 25% regular Chardonnay, and 15% Blanc de Blanc.
   - **Other production activities and costs not directly associated with a specific wine –** Allocate according to the number of bottles produced of each wine.
   - **Administrative activities –** Allocate according to sales revenues.
   a. Use this information, along with the information given in the case, to construct a product profitability analysis. *Assume that AVS did not buy the grapes referred to in Question 2.*
   b. What would you recommend to Kay Aproveche regarding the product profitability analysis?

Exhibit 1
A Votre Sante income statement -2010

Revenues	$ 848,000	
Product Costs		
Grapes	$ 124,000	
Bottle, Labels, Corks	122,500	
Harvest Labor	18,000	
Crush Labor	3,000	
Indirect Materials	6,329	
Depreciation	8,100	
Lab Expenses	8,000	
Production Office	12,000	
Utilities	5,500	
Waste Treatment	2,000	
Wine Master	15,000	
Supervisor	55,000	
Barrels	4,725	
Total Product Costs	$ 384,154	45.3%
Gross Margin	$ 463,846	54.7%
Administrative & Sales Costs		
Administrative Rent & Office	$ 20,000	
Liquor Taxes	147,000	
Sales Commissions	98,000	
Sales Manager	30,000	
Administrative Salary	75,000	
Total Administrative & Sales Costs	$ 370,000	
Operating Margin	$ 93,846	11.1%

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